

5.13.16

AI25BTECH11030 -Sarvesh Tamgade

Question: The equation of a circle with origin as centre and passing through the vertices of an equilateral triangle whose median is of length $3a$ is

- 1) $x^2 + y^2 = 9a^2$
- 2) $x^2 + y^2 = 16a^2$
- 3) $x^2 + y^2 = 4a^2$
- 4) $x^2 + y^2 = a^2$

Solution: Solution: Let the vertices of the equilateral triangle be **A**, **B**, and **C**. And the radius of circle be r .

Since the circle is centered at the origin, and the radius is r :

$$\mathbf{u} = -\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (4.1)$$

$$f = -r^2 \quad (4.2)$$

Since the circle passes through **A**:

$$\|\mathbf{x}^\top \mathbf{x}\| + 2\mathbf{u}^\top \mathbf{x} + f = 0 \quad (4.3)$$

$$\|\mathbf{x}^\top \mathbf{x}\| + 2\begin{pmatrix} 0 & 0 \end{pmatrix} \mathbf{x} + f = 0 \quad (4.4)$$

$$\|\mathbf{x}^\top \mathbf{x}\| - r^2 = 0 \quad (4.5)$$

$$(4.6)$$

For an equilateral triangle inscribed in a circle, let us place the vertices as:

$$\mathbf{A} = \begin{pmatrix} r \\ 0 \end{pmatrix} \quad (4.7)$$

$$\mathbf{B} = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} \quad (4.8)$$

$$\mathbf{C} = \begin{pmatrix} r \cos \theta \\ -r \sin \theta \end{pmatrix} \quad (4.9)$$

The median from vertex **A** goes to the midpoint of **B** and **C**:

$$\text{Midpoint} = \frac{\mathbf{B} + \mathbf{C}}{2} = \begin{pmatrix} r \cos \theta \\ 0 \end{pmatrix} \quad (4.10)$$

The median vector is:

$$\mathbf{m} = \begin{pmatrix} r \cos \theta \\ 0 \end{pmatrix} - \begin{pmatrix} r \\ 0 \end{pmatrix} = \begin{pmatrix} r(\cos \theta - 1) \\ 0 \end{pmatrix} \quad (4.11)$$

Given that the median length is $3a$:

$$\|\mathbf{m}^\top \mathbf{m}\| = (3a)^2 = 9a^2 \quad (4.12)$$

$$r^2(\cos \theta - 1)^2 = 9a^2 \quad (4.13)$$

For an equilateral triangle, $\cos \theta = \cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}$:

$$(\cos \theta - 1) = -\frac{3}{2} \quad (4.14)$$

$$r^2 \frac{9}{4} = 9a^2 \quad (4.15)$$

$$r^2 = 4a^2 \quad (4.16)$$

Therefore, the equation of the circle is:

$$\|\mathbf{x}^\top \mathbf{x}\| + 2 \begin{pmatrix} 0 & 0 \end{pmatrix} \mathbf{x} - 4a^2 = 0 \quad (4.17)$$

$$\|\mathbf{x}^\top \mathbf{x}\| - 4a^2 = 0 \quad (4.18)$$

Answer: The orthocentre of the triangle is Therefore, the equation of the circle is:

$$x^2 + y^2 = 4a^2 \quad (4.19)$$

The answer is (c).

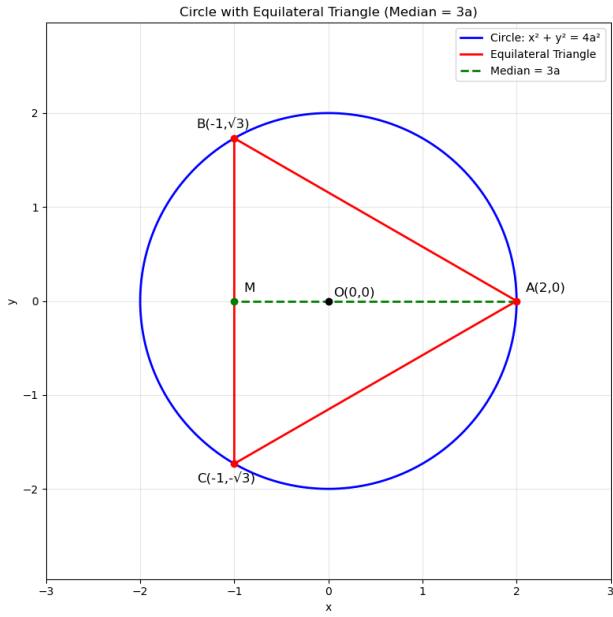


Fig. 4.1: Vector Representation